Laplace transforms for initial value problems

Now that we understand how to use the formulas for the Laplace transform of derivative functions,

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

and we know how to perform inverse Laplace transforms, we want to apply these two concepts to solving initial value problems.

Solving initial value problems

Remember that, given some second order nonhomogeneous equation ($f(t) \neq 0$) in the general form

$$ay'' + by' + cy = f(t)$$

we can find its solution using a Laplace transform, as long as we have values for the initial conditions $f(0)$ and $f'(0)$.

In order to solve this kind of initial value problem using a Laplace transform, we'll follow these steps:

1. Use formulas from the table to transform y'', y', y, and $f(t)$.
2. Plug in the initial conditions to simplify the transformation.
3. Use algebra to solve for $Y(s)$, then partial fractions to decompose it.

© Krista King Math



4. Use an inverse Laplace transform to put the solution to the second order nonhomogeneous differential equation back in terms of t, instead of s.

Let's work through an example so that we can see how these steps get applied.

Example

Use a Laplace transform to find the solution to the differential equation, given $y(0) = -1$ and $y'(0) = 1$.

$$y'' - 6y' + 5y = 5t$$

From our table of Laplace transforms, we know

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(t) = \frac{1}{s^2}$$

Plugging these transforms into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) - 6[sY(s) - y(0)] + 5Y(s) = 5\left(\frac{1}{s^2}\right)$$

© Krista King Math

Now we'll plug in the initial conditions $y(0) = -1$ and $y'(0) = 1$ in order to simplify the transform.

$$s^2Y(s) - s(-1) - 1 - 6[sY(s) - (-1)] + 5Y(s) = \frac{5}{s^2}$$

$$s^2Y(s) + s - 1 - 6[sY(s) + 1] + 5Y(s) = \frac{5}{s^2}$$

$$s^2Y(s) + s - 1 - 6sY(s) - 6 + 5Y(s) = \frac{5}{s^2}$$

$$s^2Y(s) + s - 6sY(s) + 5Y(s) - 7 = \frac{5}{s^2}$$

Solve for $Y(s)$ by collecting all the $Y(s)$ terms on one side, and moving all other terms to the other side.

$$s^2Y(s) - 6sY(s) + 5Y(s) = \frac{5}{s^2} + 7 - s$$

$$s^2Y(s) - 6sY(s) + 5Y(s) = \frac{5 + 7s^2 - s^3}{s^2}$$

Factor out $Y(s)$, then isolate it on the left side of the equation.

$$Y(s)(s^2 - 6s + 5) = \frac{5 + 7s^2 - s^3}{s^2}$$

$$Y(s)(s - 5)(s - 1) = \frac{5 + 7s^2 - s^3}{s^2}$$

$$Y(s) = \frac{5 + 7s^2 - s^3}{s^2(s - 5)(s - 1)}$$

© Krista King Math

We'll need to use a partial fractions decomposition.

$$\frac{5+7s^2-s^3}{s^2(s-5)(s-1)}=\frac{A}{s}+\frac{B}{s^2}+\frac{C}{s-5}+\frac{D}{s-1}$$

To solve for B, we'll remove the s^2 factor, set $s=0$, and find the value of the left side.

$$\frac{5+7s^2-s^3}{(s-5)(s-1)}$$

$$\frac{5+7(0)^2-0^3}{(0-5)(0-1)}=\frac{5}{(-5)(-1)}=\frac{5}{5}=1$$

To solve for C, we'll remove the $(s-5)$ factor, set $s=5$, and find the value of the left side.

$$\frac{5+7s^2-s^3}{s^2(s-1)}$$

$$\frac{5+7(5)^2-5^3}{5^2(5-1)}=\frac{5+7(25)-125}{25(4)}=\frac{5+175-125}{100}=\frac{55}{100}=\frac{11}{20}$$

To solve for D, we'll remove the $(s-1)$ factor, set $s=1$, and find the value of the left side.

$$\frac{5+7s^2-s^3}{s^2(s-5)}$$

$$\frac{5+7(1)^2-1^3}{1^2(1-5)}=\frac{5+7-1}{1-5}=-\frac{11}{4}$$

© Krista King Math



Now that we have the values of B, C, and D, we can plug those into the decomposition,

$$\frac{5+7s^2-s^3}{s^2(s-5)(s-1)} = \frac{A}{s} + \frac{1}{s^2} + \frac{\frac{11}{20}}{s-5} + \frac{-\frac{11}{4}}{s-1}$$

and then pick a value we haven't used for s yet ($s \neq 0, 1, 5$), like $s = -1$, plug it in, and solve for A.

$$\frac{5+7(-1)^2-(-1)^3}{(-1)^2(-1-5)(-1-1)} = \frac{A}{-1} + \frac{1}{(-1)^2} + \frac{\frac{11}{20}}{-1-5} + \frac{-\frac{11}{4}}{-1-1}$$

$$\frac{5+7+1}{(-6)(-2)} = -A + 1 - \frac{\frac{11}{20}}{6} + \frac{\frac{11}{4}}{2}$$

$$\frac{13}{12} = -A + 1 - \frac{11}{120} + \frac{11}{8}$$

$$A = 1 - \frac{11}{120} + \frac{11}{8} - \frac{13}{12}$$

$$A = \frac{120}{120} - \frac{11}{120} + \frac{165}{120} - \frac{130}{120}$$

$$A = \frac{144}{120}$$

$$A = \frac{6}{5}$$

Plugging the values we found for A, B, C, and D back into the partial fractions decomposition gives

© Krista King Math



$$Y(s) = \frac{\frac{6}{5}}{s} + \frac{1}{s^2} + \frac{\frac{11}{20}}{s-5} + \frac{-\frac{11}{4}}{s-1}$$

and then we can rearrange each term in the decomposition to make it easier to find a matching formula in the Laplace transform table.

$$Y(s) = \frac{6}{5}\left(\frac{1}{s}\right) + \left(\frac{1}{s^2}\right) + \frac{11}{20}\left(\frac{1}{s-5}\right) - \frac{11}{4}\left(\frac{1}{s-1}\right)$$

The terms remaining inside the parentheses should remind us of the transforms

$$\mathscr{L}(1) = \frac{1}{s} \qquad \mathscr{L}(t) = \frac{1}{s^2}$$

$$\mathscr{L}(e^{5t}) = \frac{1}{s-5} \qquad \mathscr{L}(e^{t}) = \frac{1}{s-1}$$

or equivalently, the inverse transforms

$$\mathscr{L}^{-1}\left(\frac{1}{s}\right) = 1 \qquad \mathscr{L}^{-1}\left(\frac{1}{s^2}\right) = t$$

$$\mathscr{L}^{-1}\left(\frac{1}{s-5}\right) = e^{5t} \qquad \mathscr{L}^{-1}\left(\frac{1}{s-1}\right) = e^{t}$$

So we'll make these substitutions to put the equation back in terms of the original variable t, instead of the transform variable s, which will give us the solution to the second order nonhomogeneous differential equation.

$$y(t) = \frac{6}{5} + t + \frac{11}{20}e^{5t} - \frac{11}{4}e^{t}$$

© Krista King Math

When the initial conditions aren't for $t = 0$

As we mentioned before, because $f(0)$ and $f'(0)$ are baked into the Laplace transforms of the first and second derivatives,

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

we must have initial conditions defined at $t = 0$. Therefore, if we're given initial conditions at some non-zero value of t, we'll need to shift them to $t = 0$ before we can find the solution to the second order differential equation.

This will involve making a substitution that replaces the non-zero value. Let's work through an example so that we can see how to handle this.

Example

Use a Laplace transform to find the solution to the differential equation, given $y(2) = 0$ and $y'(2) = 2$.

$$y'' - 3y' = e^{-3t}$$

Before we do anything else, we need to shift the initial conditions from $t = 2$ to $t = 0$. We'll define a new variable η, and we want η to be equal to 0

© Krista King Math



so that we can set the initial conditions at $y(\eta = 0)$ and $y'(\eta = 0)$. So η and t are related by

$$\eta = t - 2$$

$$t = \eta + 2$$

Then we want to make substitutions for t into the differential equation.

$$y'' - 3y' = e^{-3t}$$

$$y''(t) - 3y'(t) = e^{-3t}$$

$$y''(\eta + 2) - 3y'(\eta + 2) = e^{-3(\eta+2)}$$

$$y''(\eta + 2) - 3y'(\eta + 2) = e^{-3\eta-6}$$

$$y''(\eta + 2) - 3y'(\eta + 2) = e^{-6}e^{-3\eta}$$

Next, we'll define a new function $u(\eta)$ with

$$u(\eta) = y(\eta + 2)$$

$$u'(\eta) = y'(\eta + 2)$$

$$u''(\eta) = y''(\eta + 2)$$

Then the initial conditions are

$$u(0) = y(0 + 2) = y(2) = 0$$

$$u'(0) = y'(0 + 2) = y'(2) = 2$$

After all of this shifting, the new initial value problem is

© Krista King Math



$u'' - 3u' = e^{-6}e^{-3\eta}$ with $u(0) = 0$ and $u'(0) = 2$

From our table of Laplace transforms, we know

$$\mathscr{L}(u'') = s^2U(s) - su(0) - u'(0)$$

$$\mathscr{L}(u') = sU(s) - u(0)$$

$$\mathscr{L}(e^{-3\eta}) = \frac{1}{s-(-3)} = \frac{1}{s+3}$$

Plugging these transforms into the differential equation gives

$$s^2U(s) - su(0) - u'(0) - 3(sU(s) - u(0)) = e^{-6}\left(\frac{1}{s+3}\right)$$

Now we'll plug in the initial conditions $u(0) = 0$ and $u'(0) = 2$ in order to simplify the transform.

$$s^2U(s) - 2 - 3sU(s) = e^{-6}\left(\frac{1}{s+3}\right)$$

Solve for $U(s)$ by collecting all the $U(s)$ terms on one side, and moving all other terms to the other side.

$$s^2U(s) - 3sU(s) = e^{-6}\left(\frac{1}{s+3}\right) + 2$$

Factor out $U(s)$, then isolate it on the left side of the equation.

$$U(s)(s^2 - 3s) = e^{-6}\left(\frac{1}{s+3}\right) + 2$$

© Krista King Math



$$U(s) = e^{-6}\left(\frac{1}{(s+3)(s^2-3s)}\right) + \frac{2}{(s^2-3s)}$$

$$U(s) = \frac{e^{-6}}{s(s+3)(s-3)} + \frac{2}{s(s-3)}$$

$$U(s) = \frac{e^{-6}}{s(s+3)(s-3)} + \frac{2(s+3)}{s(s+3)(s-3)}$$

$$U(s) = \frac{e^{-6}+2(s+3)}{s(s+3)(s-3)}$$

We'll need to use a partial fractions decomposition.

$$\frac{e^{-6}+2(s+3)}{s(s+3)(s-3)} = \frac{A}{s} + \frac{B}{s+3} + \frac{C}{s-3}$$

To solve for A, we'll remove the s factor, set $s = 0$, and find the value of the left side.

$$\frac{e^{-6}+2(s+3)}{(s+3)(s-3)}$$

$$\frac{e^{-6}+2(0+3)}{(0+3)(0-3)} = \frac{e^{-6}+2(3)}{(3)(-3)} = -\frac{e^{-6}+6}{9}$$

To solve for B, we'll remove the $(s+3)$ factor, set $s = -3$, and find the value of the left side.

$$\frac{e^{-6}+2(s+3)}{s(s-3)}$$

© Krista King Math

$$\frac{e^{-6}+2(-3+3)}{-3(-3-3)}=\frac{e^{-6}+2(0)}{-3(-6)}=\frac{e^{-6}}{18}$$

To solve for C, we'll remove the $(s-3)$ factor, set $s=3$, and find the value of the left side.

$$\frac{e^{-6}+2(s+3)}{s(s+3)}$$

$$\frac{e^{-6}+2(3+3)}{3(3+3)}=\frac{e^{-6}+2(6)}{3(6)}=\frac{e^{-6}+12}{18}$$

Plugging the values we found for A, B, and C back into the partial fractions decomposition gives

$$U(s)=\frac{-\frac{e^{-6}+6}{9}}{s}+\frac{\frac{e^{-6}}{18}}{s+3}+\frac{\frac{e^{-6}+12}{18}}{s-3}$$

and then we can rearrange each term in the decomposition to make it easier to find a matching formula in the Laplace transform table.

$$U(s)=-\frac{e^{-6}+6}{9}\left(\frac{1}{s}\right)+\frac{e^{-6}}{18}\left(\frac{1}{s+3}\right)+\frac{e^{-6}+12}{18}\left(\frac{1}{s-3}\right)$$

The terms remaining inside the parentheses should remind us of the transforms

$$\mathscr{L}(1)=\frac{1}{s}\qquad\mathscr{L}(e^{3\eta})=\frac{1}{s-3}\qquad\mathscr{L}(e^{-3\eta})=\frac{1}{s+3}$$

or equivalently, the inverse transforms

© Krista King Math

$$\mathscr{L}^{-1}\left(\frac{1}{s}\right) = 1 \qquad \mathscr{L}^{-1}\left(\frac{1}{s-3}\right) = e^{3\eta} \qquad \mathscr{L}^{-1}\left(\frac{1}{s+3}\right) = e^{-3\eta}$$

So we'll make these substitutions to put the equation back in terms of η, instead of the transform variable s,

$$u(\eta) = -\frac{e^{-6}+6}{9} + \frac{e^{-6}}{18}e^{-3\eta} + \frac{e^{-6}+12}{18}e^{3\eta}$$

$$u(\eta) = -\frac{e^{-6}}{9} - \frac{6}{9} + \frac{e^{-6}}{18}e^{-3\eta} + \frac{e^{-6}}{18}e^{3\eta} + \frac{12}{18}e^{3\eta}$$

$$u(\eta) = -\frac{1}{9e^6} - \frac{2}{3} + \frac{1}{18e^6e^{3\eta}} + \frac{e^{3\eta}}{18e^6} + \frac{2e^{3\eta}}{3}$$

Because $y(t) = u(\eta) = u(t-2)$ and $\eta = t-2$, the solution to the original differential equation can be written as

$$y(t) = -\frac{1}{9e^6} - \frac{2}{3} + \frac{1}{18e^6e^{3(t-2)}} + \frac{e^{3(t-2)}}{18e^6} + \frac{2e^{3(t-2)}}{3}$$

$$y(t) = -\frac{1}{9e^6} - \frac{2}{3} + \frac{1}{18e^6e^{3t-6}} + \frac{e^{3t-6}}{18e^6} + \frac{2e^{3t-6}}{3}$$

$$y(t) = -\frac{2}{3} + \frac{1}{18}e^{-3t} - \frac{1}{9}e^{-6} + \frac{2}{3}e^{3t-6} + \frac{1}{18}e^{3t-12}$$

© Krista King Math